

Electromagnetic Theory (EE3005)

Date: September 24, 2024

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Roll No

Section

Stu

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1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 01: Demonstrate the use of 3D orthogonal coordinate system and vector analysis tools (curl, divergence, etc.) in problem solving

Q1: The Fig.1 shown ahead is a 3D image drawn using values $0 \leq r \leq 1, 0 \leq \theta \leq \frac{\pi}{4}, 0 \leq \phi \leq \frac{7\pi}{4}$ (solid) [20 marks]

image drawn using values
(mentioned at the top of the figure)
from spherical coordinate system
(SCS). Using differential method,
find.

- (a) the total volume of the image.
- (b) the total surface area of the image.
- (c) the length of the circular edge as mentioned in the Fig. 1.

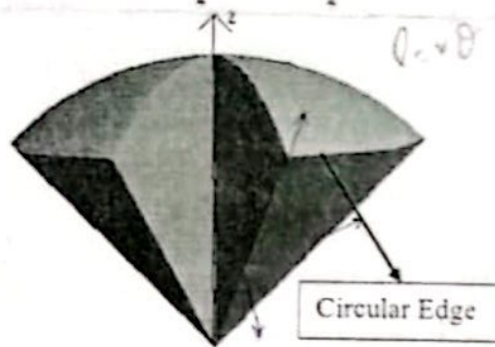


Figure 1. A 3D image with circular edge

- (d) Also, convert the point $P(r = 1\text{m}, \theta = \frac{\pi}{4}, \phi = \frac{7\pi}{4})$ into RCS.

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \tan^{-1}(y/z) \\ \phi &= \tan^{-1}(y/x) \end{aligned}$$

$$r \sin \theta = \rho$$

$$r \cos \theta = z$$

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q2: A uniform line charge is distributed along the z -axis from $z = 5\text{m}$ to ∞ and from $z = -5\text{m}$ to $-\infty$ as shown in the Fig. 2 with $\rho_L = 20 \text{ [nC/m]}$. There is no line charge from $z = -5$ to 5m .

[20 marks]

Formulate the problem for $d\vec{E}$ in RCS and compute the total electric field intensity, $\vec{E}$, at $(2, 0, 0)\text{m}$ in CCS.

Hint: Finding symmetry in the problem may reduce the components you need to find.

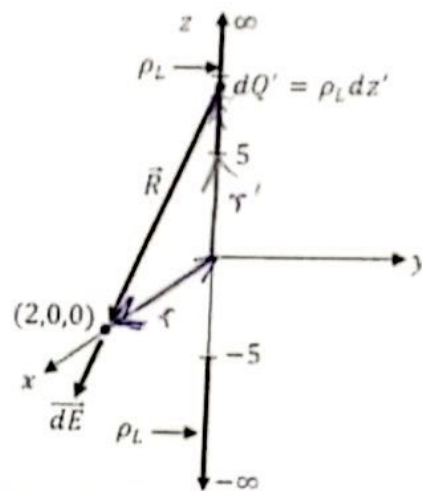


Figure 2 Two semi-infinite line charges on z -axis

$$\frac{\rho_L}{2\pi\epsilon_0 r}$$

Hint:

$$\int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

and

$$\int \frac{x dx}{(a^2 + x^2)^{3/2}} = \frac{-1}{\sqrt{a^2 + x^2}}$$

d $\times 10^{-1}$
C $\times 10^{-2}$
m $\times 10^{-3}$
 μ $\times 10^{-6}$
n $\times 10^{-9}$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{F}{Q}$$